

Facebook's MyRocks Truly Rocks!

MyRocks is a unique database engine designed by Facebook to work with RocksDB so that MySQL can be used with the database. Even though InnoDB (another DB in use at Facebook) seems better than MyRocks in certain ways, in the case of the former, MySQL cannot be used with an embedded database. Therefore, MyRocks is considered better than InnoDB and it really rocks!



As data is the basic requirement of any software system, storing data in a safe and easily accessible manner is the biggest challenge. That's where a database comes into play. It can organise, store and retrieve data from a computer storage system using software. Such software is known as a database management system (DBMS), and it helps users and applications to store or retrieve data from it. However, when the database is too large and on a cloud network or server, then it becomes tedious to store and retrieve data effectively. That's why a database

engine is needed with a DBMS to perform database operations with minimum latency.

Various DBMSs like MySQL, PostgreSQL, MSSQL, Oracle Database and Microsoft Access can be used with a database server. Such DBMSs were being used with database servers that had traditional magnetic hard drives. However, magnetic drives can cause a lot of latency compared to a solid state drive (SSD). As SSDs use Flash memory chips instead of a magnetic disk or any mechanical moving parts, the read/write speed of SSDs is faster than traditional magnetic drives. That's why such SSDs can be used in a database server to store/

retrieve data with minimum latency. So, for SSD technology with a database server, special database software is required, and that's where RocksDB enters the picture!

With RocksDB, it's possible to use SSDs in a database server. It's an embedded database that can be used with SSDs and requires very little maintenance. It was created by Facebook in May 2012, using the C++ programming language, in order to overcome the disadvantages of InnoDB, which caused read/write/space amplification factors. However, Facebook wanted MySQL features with the database servers, using SSDs. As MySQL is great for backup and automation, Facebook wanted to use it along with RocksDB. That's why Facebook developed the MyRocks database engine in 2015 and it was also written in C++. MyRocks integrates RocksDB and MySQL in a way that SQL can be used by Facebook applications with the RocksDB database server. MyRocks is an open source database engine that helps to store/retrieve data from an embedded database like RocksDB, using MySQL.

Requirements and limitations of MyRocks

- MyRocks requires C++ compilers and libraries for development.
- It works with RocksDB to implement the MySQL layer.
- It is based on Oracle MySQL 5.6.
- MyRocks is included from MariaDB 10.2.5 onwards.

